IDENTIFICATION OF LITHOLOGY FROM WELL LOG DATA USING MACHINE LEARNING

OBJECTIVES: THIS STUDY EMPLOYS MULTIPLE MACHINE LEARNING MODELS TO DISCERN LITHOLOGY FROM THE WELL-LOG DATA OF THE HUGOTON AND PANOMA FIELDS (NORTH AMERICA).

INTRODUCTION: Reservoir characterisation and geomechanical modelling benefit significantly from diverse machine learning techniques, addressing complexities inherent in subsurface information. Accurate lithology identification is pivotal, furnishing crucial insights into subsurface geological formations. Lithology is pivotal in appraising hydrocarbon accumulation potential and optimising drilling strategies.

METHODS: The well log data of the Hugoton and Panoma Fields(North America) comprises over 3000 data points with diverse features influencing the target variable, lithology. The dataset encompasses four primary lithologies— sandstone, limestone, marl, and claystone— constituting a complex subsurface stratum. Lithology identification is framed as a classification problem, and four distinct ML algorithms are deployed to train and assess the models, partitioning the dataset into a 8:2 ratio for training and testing, respectively.

RESULTS: The resulting confusion matrix indicates a close alignment between predicted and true labels. While all algorithms exhibit favourable performance, the Neural Network algorithm demonstrates the highest efficacy, yielding an exceptional overall accuracy of 0.91.

CONCLUSION: Notably, this model's training spans diverse wells within the same basin, showcasing its capability to predict lithology within intricate strata. Additionally, its robustness positions it as a potential tool for identifying other properties of rock formations.

PROPOSED METHODOLGY

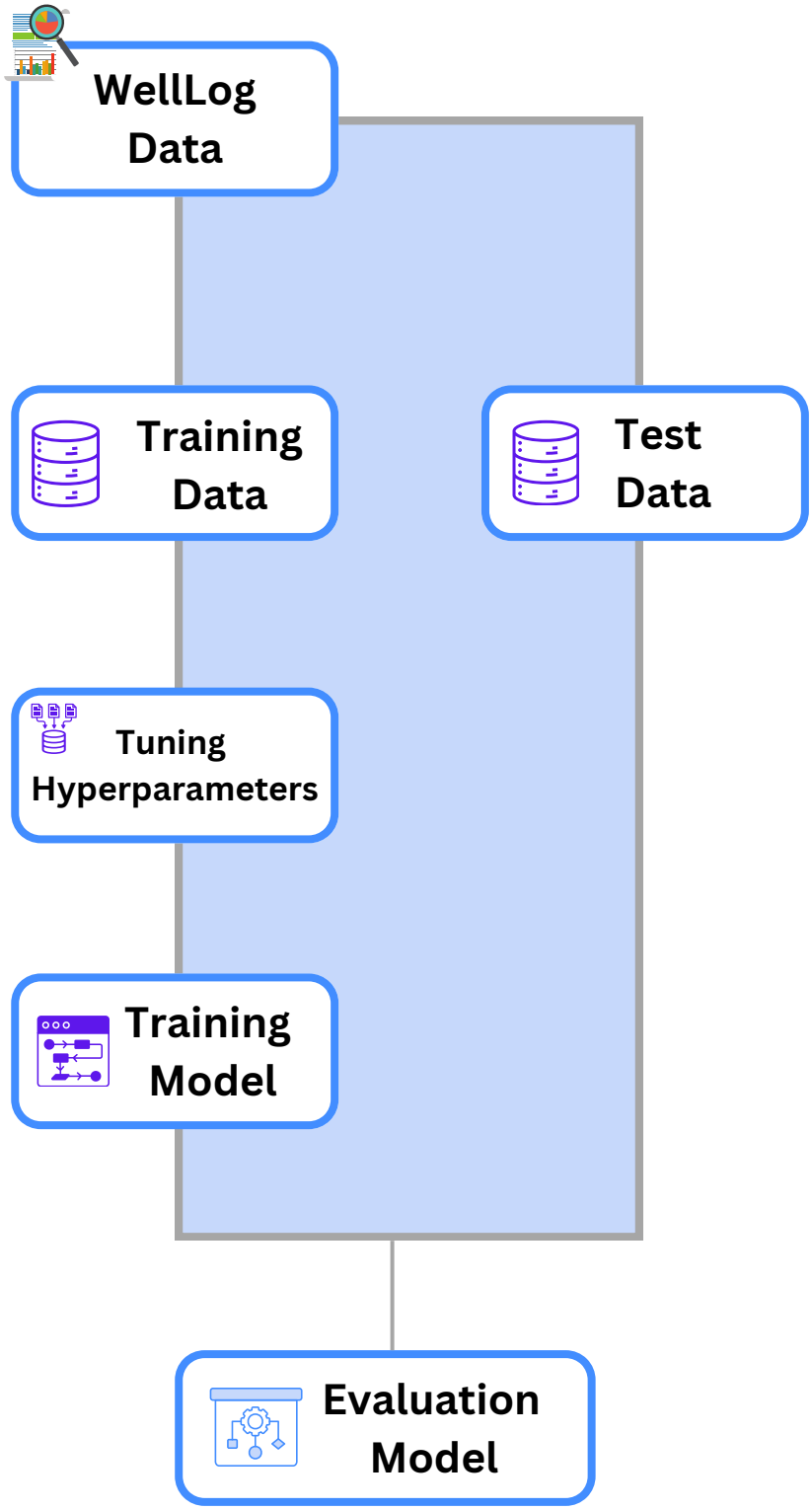


Figure 1. Methodology Process for the Identification of Lithology from Well Log Data using Machine Learning

MACHINE LEARNING MODELS

RANDOM FOREST CLASSIFIER -

The Random Forest classifier is an ensemble learning technique that combines multiple decision trees to create a robust and accurate model. It utilizes the bagging method, specifically bootstrap aggregating, to construct numerous decision trees. In this process, each tree is trained using a random subset of both features and data samples from the available dataset. The integration of predictions from all the trees allows the Random Forest to produce a final prediction. In classification tasks, this prediction is based on the consensus or majority vote among the individual trees.

XGBOOST CLASSIFIER -

It is an implementation of gradient boosting that builds an ensemble of decision trees to improve predictive performance. XGBoost stands for Extreme Gradient Boosting, and it is known for its efficiency, scalability, and performance. It utilizes advanced techniques like regularization to prevent overfitting and employs parallel processing to speed up computation, making it a popular choice for large and complex datasets.

DECISION TREES CLASSIFIER -

It models data by splitting it into branches based on feature values, forming a tree-like structure where each node represents a decision rule, and each leaf node represents a class label. The tree is built through a process of recursive partitioning, selecting splits that best separate the data according to a criterion like Gini impurity or information gain.

NEURAL NETWORK -

A Neural Network Classifier is a machine learning model inspired by the structure and function of the human brain. It consists of interconnected layers of nodes (neurons), including an input layer, one or more hidden layers, and an output layer. Each neuron processes input data using weights and activation functions to produce an output. Neural networks excel at capturing complex patterns and relationships in data.

DATASET

ORIGIN:
THE DATASET COMES FROM A CLASS EXCERCISE FROM THE UNIVERSITY OF KANSAS AND CONTAINS LOGS FROM THE LARGEST GAS FIELDS IN NORTH AMERICA, THE HUGOTON AND PANOMA FIELDS.

THE TARGET VARIABLE -- FACIES

Facies (types of rocks) are studied from core samples in every half foot and matched with logging data in well location.

THE NINE DISCRETE FACIES (CLASSES OF ROCKS) ARE:

- **SS** : Nonmarine sandstone
- **CSiS** : Nonmarine coarse siltstone
- **FSiS** : Nonmarine fine siltstone
- **SiSH** : Marine siltstone and shale
- **MS** : Mudstone (limestone)
- **WS** : Wackestone (limestone)
- **D** : Dolomite
- **PS** : Packstone-grainstone (limestone)
- **BS** : Phylloid-algal bafflestone (limestone)

LOGGING DATA PROVIDED ARE --

Feature variables include five from wireline log measurements and two geologic constraining variables that are derived from geologic knowledge.

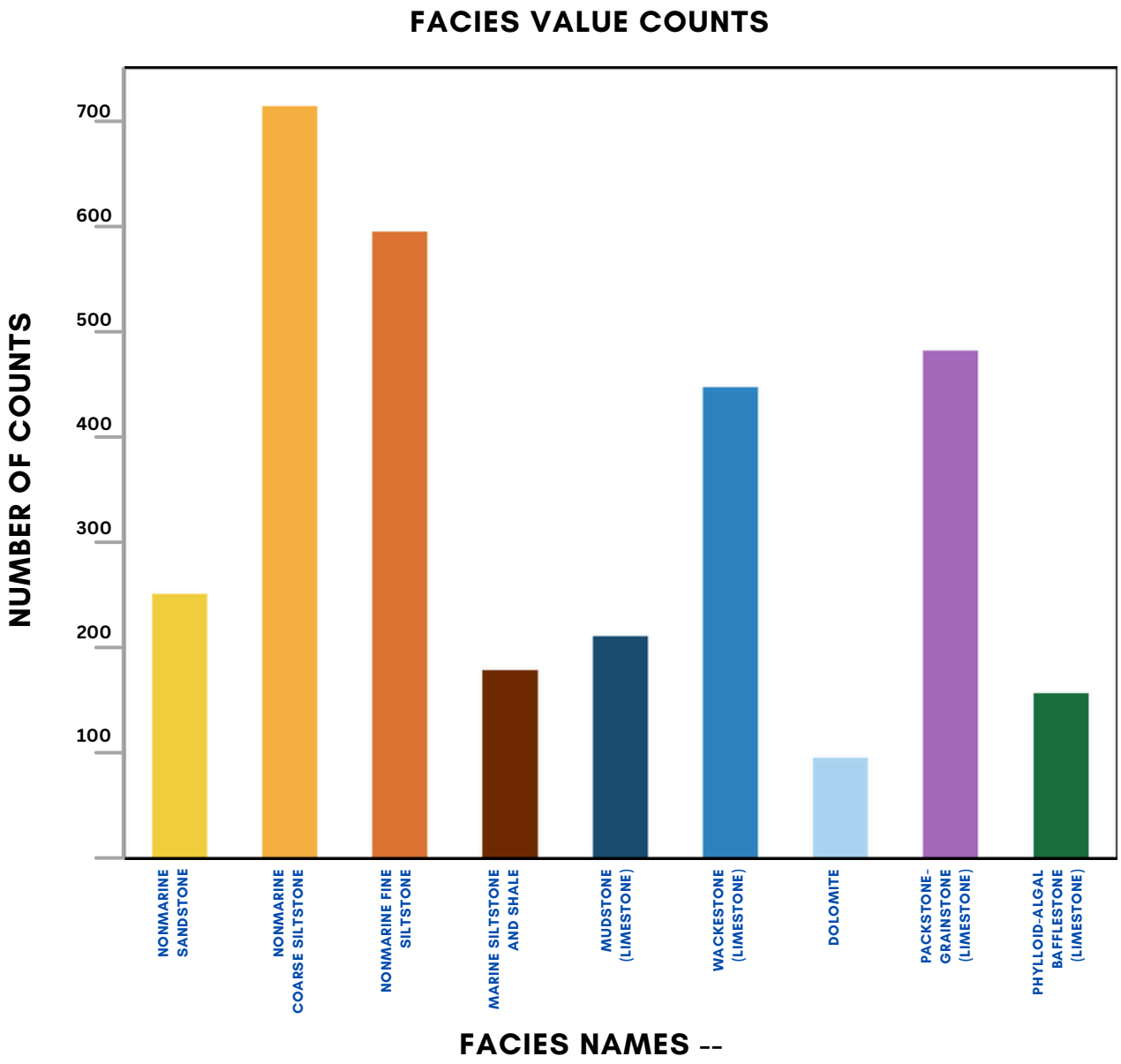
- **Gamma Ray** : this wireline logging tools measure gamma emission
- **ILD_log10** : this is resistivity measurement
- **PE** : photoelectric effect log
- **DeltaPHI** : Phi is a porosity index in petrophysics.
- **PNHIND** : Average of neutron and density log.
- **NM_M** :nonmarine-marine indicator
- **RELPOS** : relative position

DATA_OVERVIEW

DATASET --

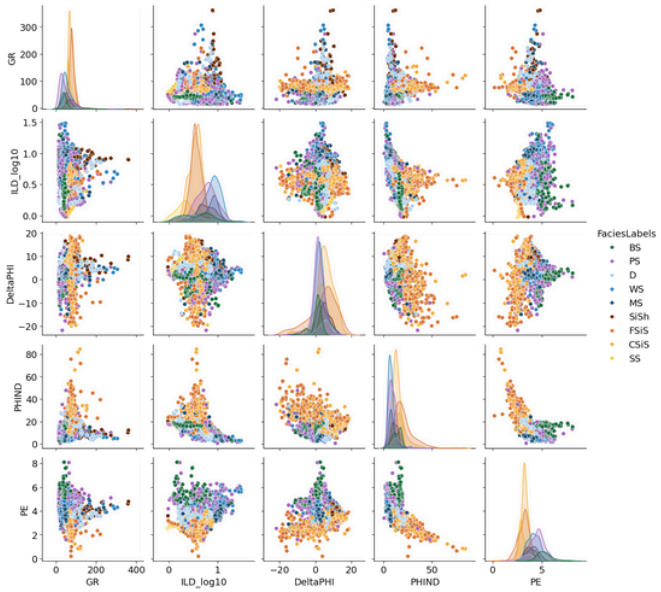
Facies	Formation	Well Name	Depth	GR	ILD_log10	DeltaPHI	PHIND	PE	NM_M	RELPOS
3	A1 SH	SHRIMPLIN	2793	77.45	0.664	9.9	11.915	4.6	1	1
3	A1 SH	SHRIMPLIN	2793.5	78.26	0.661	14.2	12.565	4.1	1	0.979
3	A1 SH	SHRIMPLIN	2794	79.05	0.658	14.8	13.05	3.6	1	0.957
3	A1 SH	SHRIMPLIN	2794.5	86.1	0.655	13.9	13.115	3.5	1	0.936
3	A1 SH	SHRIMPLIN	2795	74.58	0.647	13.5	13.3	3.4	1	0.915
3	A1 SH	SHRIMPLIN	2795.5	73.97	0.636	14	13.385	3.6	1	0.894
3	A1 SH	SHRIMPLIN	2796	73.72	0.63	15.6	13.93	3.7	1	0.872

COUNT PLOTS OF TARGET VALUES --

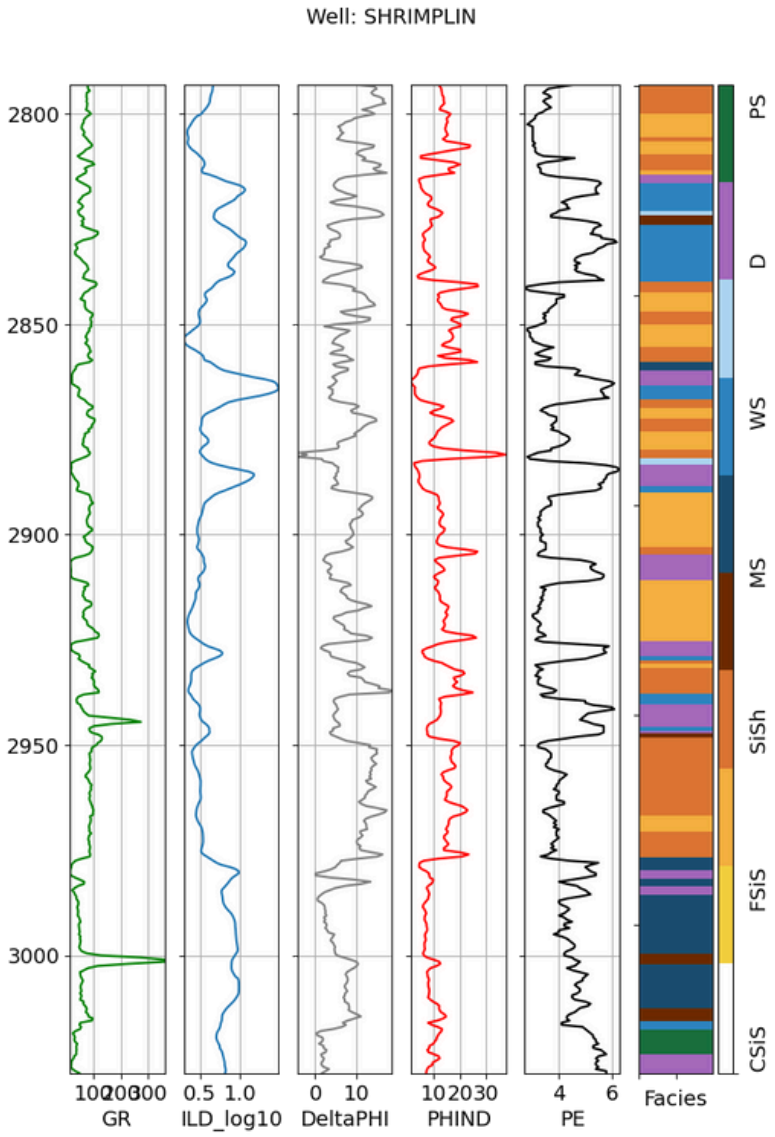


EXPLORATORY DATA ANALYSIS

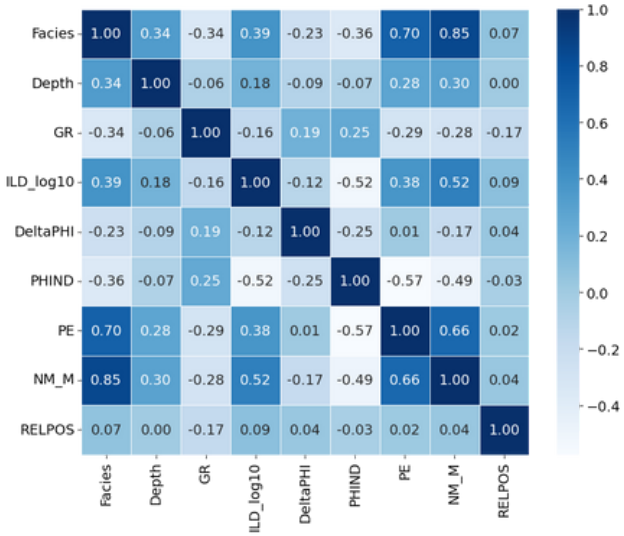
PAIRPLOT OF FACIES WITH ACCORDANCE TO VARIOUS LOG_DATA --



WELL_LOG DATA PLOTING FOR WELL - SHRIMPLIN --



PEARSON_CORRELATION MATRIX OF FACIES WITH ACCORDANCE TO VARIOUS LOG_DATA --

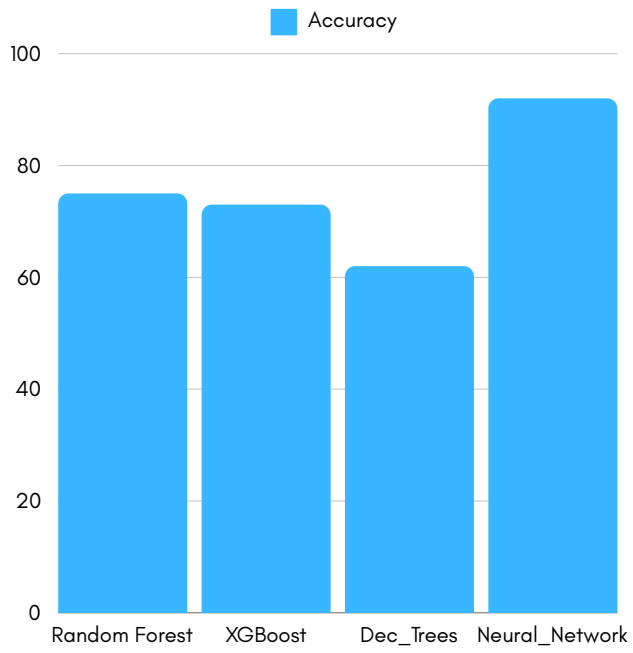


ML MODELS TESTING

CLASSIFICATION REPORT OF SELECTED ML MODELS

ML MODELS --	Random Forest Classifier	XGBoost Classifier	Decision Trees Classifier	Neural Network Classifier
ACCURACY	**74.53**	**72.68**	**61.64**	**91.53**
PRECISION	74.91	73.06	61.72	77.65
RECALL	74.51	72.66	61.64	76.97
F1_SCORE	74.34	72.48	61.58	76.85

ACCURACY SCORE OF SELECTED ML MODELS --



CONCLUSION & DISCUSSION

CONCLUSION --

The study highlights the potency of machine learning in identifying lithology from well-log data, underscoring the significance of data analysis and visualization in deciphering intricate oil and gas datasets. Leveraging these methodologies enables operators to optimize production, curtail expenses, and enhance safety protocols. Further exploration and utilization of these tools in analyzing well-log data promise augmented efficiencies and superior outcomes in the oil and gas domain.

Moreover, this implementation shows promise in elevating the accuracy and consistency of lithology identification, subsequently refining decision-making processes in hydrocarbon resource exploration and production. Overall, the study's findings emphasize the transformative impact of machine learning in revolutionizing geological analysis within the oil and gas industry.

DISCUSSION --

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Further exploration and utilization of these tools in analyzing well-log data promise augmented efficiencies and superior outcomes in the oil and gas domain. Moreover, this implementation shows promise in elevating the accuracy and consistency of lithology identification, subsequently refining decision-making processes in hydrocarbon resource exploration and production.